GGG-P-61a
July 6, 1966
SUPERSEDING
Fed. Spec. GGG-P-61
August 12, 1955

FEDERAL SPECIFICATION

PARALLELS; STEEL AND CAST-IRON, PRECISION-GROUND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers precision-ground, hardened steel, and cast-iron parallels for use in supporting work in inspection, layout, and general machine-shop practice.

1.1.1 *Federal specification coverage.* This specification does not cover all types of parallels, but only those used most commonly by the Federal Government.

1.2 Classification.

1.2.1 *Types.* Parallels covered by this specification shall be of the following types, as specified (see 6.2):

I—Solid.
II—Ribbed.
III—Box.
IV—Adjustable.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

TT-E-489—Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Standards Association, Inc. (ASA) Standard:
B46.1—Surface Texture.

(Application for copies should be addressed to the American Standards Association, 10 East 40th St., New York, N. Y., 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of parallels which are otherwise in accordance with the requirements of this specification.

3.2 **Material.** The material used in the fabrication of the parallels shall be new and of the quality necessary to produce parallels to meet the requirements specified herein.

3.3 **Surfaces.** Surfaces of parallels other than working surfaces shall be cleaned and free of sand, dirt, fins, sprues, scales, and other harmful or extraneous materials.

3.4 **Marking for identification.** Parallels shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.5 **Sets or pairs.** When sets or pairs of parallels are specified (see 6.2), they shall be matched within the tolerance requirements specified for the type of parallel to be furnished. Each set or pair shall be furnished in plastic or fabric wraps or rolls or in metal or wooden boxes as specified (see 6.2). Wraps or rolls shall be of a durable plastic or fabric with individual pockets for each parallel, equipped with snaps or ties. Plastic materials shall be of the slow burning type.

3.6 **Boxes.** When boxes are specified for sets (see 6.2), they shall be either wood, metal, or plastic as specified (see 6.2), with partitions for holding the parallels and a cover having a positive or friction type catch. The interior and exterior surfaces of wood and metal boxes shall have a protective coating of paint or other appropriate material. Plastic materials shall be of the slow burning type.

3.7 **Type I, solid.** Type I parallels shall be of hardened steel, rectangular in cross section, when specified (see 6.2), they shall be furnished singly or in pairs.

3.7.1 *Heat treatment.* Parallels shall be heat treated to a Rockwell hardness of from 55 to 60 on the C scale, be free from cracks, and shall be stable after the final heat treatment (see 4.4.3).

3.7.2 *Finish.* Parallels shall be finish ground on all four sides to 8 microinches roughness height rating or better. All sharp edges shall be removed and all ground surfaces shall be free from grind chatter marks, burns, and cracks.

3.7.3 *Tolerances.* Each surface of the four working surfaces shall not vary from a true plane (straightness) by more than 0.0002 inch per foot. Parallelism between opposite sides shall be as shown in table II. Adjacent sides shall be square with each other within 0.0005 inch. Type I parallels shall be similar to figure 1, conform to table I, and the tolerances in table II.

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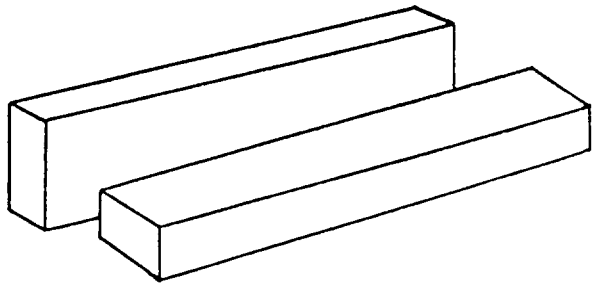


FIGURE 1. Type I, solid

TABLE I. *Type I, solid*

Width and height	Length ± 0.020
Inches	*Inches*
1/8 by 1	6
1/8 by 1-3/16	6
3/16 by 7/8	6
3/16 by 1-1/8	6
1/4 by 3/8	6
1/4 by 1/2	6
1/4 by 5/8	6
1/4 by 3/4	6
1/4 by 1	6
3/8 by 1/2	6
3/8 by 3/4	6
3/8 by 7/8	6
3/8 by 1	6
7/16 by 7/8	6
1/2 by 5/8	6
1/2 by 3/4	6
1/2 by 13/16	6
1/2 by 1	6
1/4 by 3/8	9
1/2 by 5/8	9
1/2 by 3/4	9
1/2 by 1	9
1/2 by 1-1/4	9
1/2 by 1-1/2	9
3/4 by 1	9
3/4 by 1-1/2	9
3/8 by 1	12
1/2 by 1-1/2	12
1/2 by 2	12
11/16 by 1-1/4	12
3/4 by 1	12
3/4 by 2	12
1 by 1-1/2	12
1 by 2	12
1 by 3	12
1-1/4 by 1-3/4	12
1-1/4 by 2-1/2	12
1-1/2 by 2	12
1-1/2 by 3	12

TABLE II. *Tolerances, for type I, solid*

Size	Tolerances		Width and height variation in pairs (max.)
	Width and height	Parallelism	
Inches	*Inch*	*Inch*	*Inch*
Up to 1-1/8 .	0.0002	0.0001	0.0002
1-1/8 to 1-1/2	.0003	.00015	.0003
1-1/2 to 3, inclusive ..	.0004	.0002	.0004

3.8 Type II, ribbed. Type II parallels shall be rectangular in cross section, of cast gray iron ribbed in accordance with the size for lightness and yet maintain strength and stability. The ribbing shall extend to the outer surfaces of the sides and form a part of the working surfaces.

3.8.1 *Seasoning.* After being cast and rough machined, each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure maximum stability after finishing (see 4.4.3). The finished parallels shall have a Brinell hardness of not less than 180 when using a 10-millimeter (mm.) hardened steel ball with a 3000-kilogram (kg.) load.

3.8.2 *Finish.* Type II parallels shall be finish ground on all four sides to a maximum value of 16 microinches roughness height rating. All sharp edges shall be removed and all ground surfaces shall be free of grind chatter marks and burns. All unmachined surfaces shall be finished with two coats of gray enamel conforming to TT-E-489.

3.8.3 *Tolerances.* Each surface of the four working surfaces shall not vary from a true plane (straightness) by more than 0.0002 inch per foot. Opposite sides and edges shall be parallel with each other and adjacent sides and edges shall be square with each other within 0.0005 inch in the full length of the parallel. Type II parallels shall be similar to figure 2 and conform to table III.

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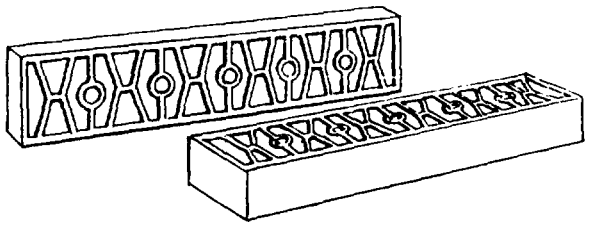


FIGURE 2. Type II, ribbed

TABLE III. *Type II, ribbed*

Length ± 0.020	Width ± 0.0005	Height ± 0.0005	Weight ± 15 Percent
Inches	*Inches*	*Inches*	*Pounds*
24	1.5	3	17
24	2	4	31
24	2.5	5	42
36	3	6	82
36	4	8	120

3.9 Type III, cross-section box. Type III parallels shall be square or rectangular in cross section, of cast gray iron. The rectangular parallels shall have a rib running the entire length to support the two larger surfaces.

3.9.1 *Seasoning.* After being cast and rough machined, each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure stability after finishing (see 4.4.3). The finished parallels shall have a Brinell hardness of not less than 180 when using a 10-mm. hardened steel ball with a 3000-kg. load.

3.9.2 *Finish.* Type III parallels shall be finish ground on all four sides and the two ends to a maximum value of 16 microinches roughness height rating. All sharp edges shall be removed and all the ground surfaces shall be free of grind chatter marks and burns.

3.9.3 *Tolerances.* Each of the six working surfaces shall not vary from a true plane (straightness) by more than 0.0002 inch per foot. The opposite sides and ends shall be parallel with each other and the adjacent sides and ends shall be square with each other within 0.0005 inch. The size shall be as specified within 0.0005 inch.

3.9.4 Type III parallels shall conform to table IV and be similar to figure 3.

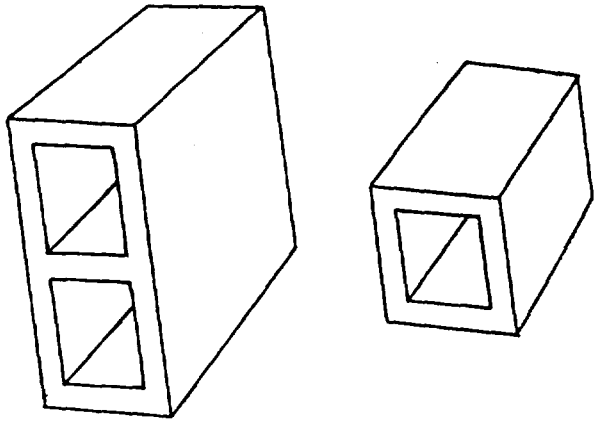


FIGURE 3. Type III, box

TABLE IV. *Type III, box*

Size	Weight ± 15 percent
Inches	*Pounds*
4 by 4 by 6	14
4 by 6 by 6	22
5 by 8 by 12	55
10 by 10 by 10	150

3.10 Type IV adjustable. Type IV parallels shall be of close-grain, thoroughly seasoned cast iron. Each parallel shall consist of two pieces, a fixed member, and a sliding member. A dovetail on the sliding member shall fit and slide smoothly in a dovetail slot in the fixed member for adjustment purposes. The total width of bearing surface between the two members shall be approximately one half of the total thickness of the parallels. The two members shall be held firmly in desired adjustment by means of a screw lock arrangement, using the number of screws specified in table V (see 6.2). They shall be supplied in individual sizes or in a complete set.

3.10.1 *Seasoning.* After being cast and rough machined, both members of each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure stability after finishing (see 4.4.3). The finished parallels shall have a hardness of not less than 87 on the Rockwell B scale.

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3.10.2 *Finish.* Type IV parallels shall be finish ground to a maximum value of 16 microinches roughness height rating on the working surfaces and 32 microinches roughness height rating on the sides.

3.10.3 *Tolerances.* Each working surface shall not vary from a true plane (straightness) by more than 0.0003 inch per foot. The top and bottom surfaces shall be parallel as specified in table V at any point within the adjustment range. Squareness of measuring faces to sides shall be within 0.001 inch per inch.

3.10.4 Type IV parallels shall be similar to figure 4 and conform to table V.

FIGURE 4. Type IV, adjustable

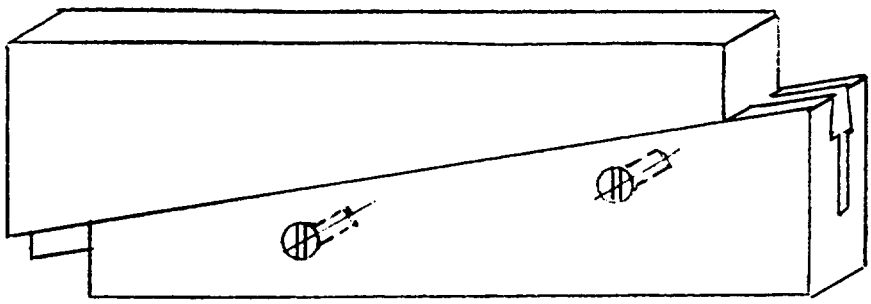


TABLE V. *Type IV, adjustable*

Size (range)	Length	Thickness	Parallelism tolerance	Number of lock screws
Inches	*Inches*	*Inch*	*Inch*	
3/8—1/2	1-3/4	9/32	0.0005	1
1/2—11/16	2-1/8	9/32	.0005	1
11/16—15/16	2-11/16	9/32	.0007	1
15/16—1-5/16	3-9/16	9/32	.0007	2
1-5/16—1-3/4	4-3/16	9/32	.0009	2
1-3/4—2-1/4	5-1/16	9/32	.0009	2

3.11 Workmanship. Workmanship shall be of the quality prevailing among manufacturers normally producing parallels of the types specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that material and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table VI.

TABLE VI. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	1.5	Percent defective	4.3.1
Dimensional and weight examination	1 ea.	II	1.5	Percent defective	4.3.2
Testing	1 ea.	II	2.5	Percent defective	4.4
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

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4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, workmanship, and marking. Defects are listed in table VII.

TABLE VII. *Classification of defects*

Categories	Defects
Critical:	None defined.
Major:	
101	Parallel not of the type and size specified.
102	Castings (types II, III, and IV) not of uniform quality, free of blowholes, cracks, or other injurious defects.
103	Working surfaces not smoothly and evenly finished.
104	Parallels not of the configuration specified.
105	Surfaces not required to have machine finish; not protected by enamel as specified.
106	Adjustable parallels (type IV) not equipped with locking screws, as specified.

4.3.2 *Dimensional and weight examination.* Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.3.

4.4.1 *Hardness test.* Each of the sample parallels shall be hardness tested in accordance with methods 242.1 or 243.1, as applicable of Fed. Test Method Std. No. 151 to determine campliance with the hardness requirements specified herein. Conversion of hardness values shall be in accordance with method 241.1 of Fed Test Method Std. No. 151.

4.4.2 *Surface finish test.* Each of the sample parallels shall be tested for surface finish in accordance with ASA B46.1 to determine compliance with the surface finish requirements specified.

4.4.3 *Stabilizing and seasoning test.* Certificate of compliance describing the methods used, the duration of the processes, and other pertinent data is required for the stabilizing and seasoning requirements of 3.7.1, 3.8.1, 3.9.1, and 3.10.1.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table VIII. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers fully prepared for delivery. lot.

TABLE VIII. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size; location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
[redacted]	Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2), the parallels shall be preserved, packaged, and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C and the level of packing A, B, or C, as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table IX is considered as part of, and used in lieu of, that portion of table I of MIL-H-15424 covering parallels.

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TABLE IX.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7
Parallels:						
Solid	P-8	IA-8				
Ribbed	P-8	IA-8				
Box	P-8	IA-8				
Adjustable	P-8	IA-8				
Up through and including 6 pounds			1		12	12
7 pounds through 30 pounds			1		2	2
31 pounds through 150 pounds			1		1	1
Set:	P-8	IA-14	1		1	1

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Types I, II, and III parallels are intended to be used for inspection and layout work or for set-ups on various machine tools.

6.1.2 Type IV parallels are useful as gages in checking the size of slots and openings and can also be used for layout work or set-ups on various machine tools.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type of parallels required (see 1.2).

(*c*) Size required (see tables I, III, IV, and V).

(*d*) Whether sets are to be furnished in plastic or fabric wraps or rolls, or metal, plastic or wood boxes (see 3.5 and 3.6).

(*e*) Whether parallels are to be furnished singly, in pairs, or in sets (see 3.7 and 3.10).

(*f*) Selection of applicable levels of packaging and packing required (see 5.1).

MILITARY CUSTODIANS:

Army—WC
Navy—SH
Air Force—67

Review activities:

Army—WC
Navy—SH
Air Force—67

User activity:

Navy—MC

Preparing activity:

Navy—SH

Orders for this specification are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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